

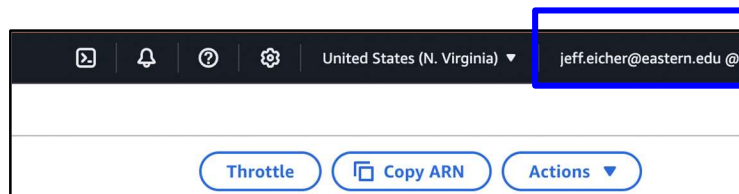


DTSC 671: AI Solutions in the Cloud

SUBMISSION TEMPLATE: Recommendation System Prototype with Amazon Personalize

Select *File > Make a copy* to edit the template and add your screenshots. Download the completed template as a PDF document to submit to Brightspace.

In every screenshot, include your AWS username that appears in the upper right corner. Example:

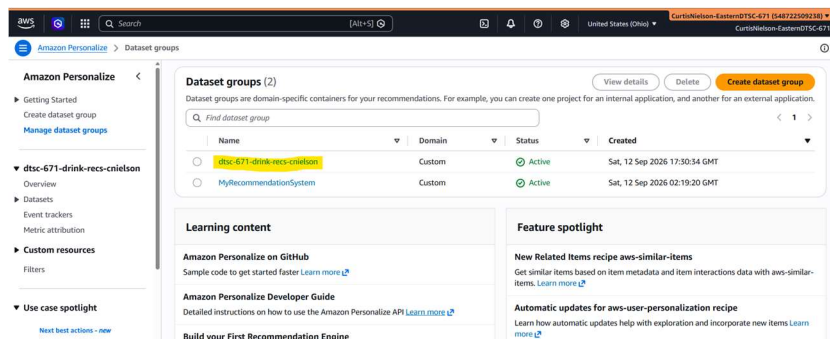


If any one screenshot is missing this identification, the entire assignment will be scored no higher than 70%.

Points totals are provided at the start of each question. You score either full credit or 0 points; there is no partial credit for each question.

Curtis Nielson
DTSC-671

- 1) [20 points] In the space below, provide a screenshot of your dataset group **dtsc-671-drink-recs-<yourname>** in Amazon Personalize.



This is course material that was distributed to students enrolled in DTSC 671 at Eastern University. The materials were made exclusively available to students solely for the purpose of completing the course and are not intended for further distribution or retention beyond the course period. The materials were created by program faculty specifically for use in courses at Eastern University.

2) [30 points] In the space below, provide a screenshot of the active *aws-user-personalization-v2* solution.

The screenshot displays the Amazon Personalize console interface. The top navigation bar shows the AWS logo, a search bar, and the account name 'CurtisNielson-EasternDTSC-671 (548722509238)' in the 'United States (Ohio)' region. The breadcrumb trail indicates the path: Amazon Personalize > Dataset groups > dtsc-671-drink-recs-cnielson > Solutions and recipes > drink-solution-cnielson.

The main content area is titled 'drink-solution-cnielson' and includes buttons for 'Clone solution', 'Update', and 'Delete'. The 'Solution overview' section provides details about the solution:

- Solution name:** drink-solution-cnielson
- Recipe:** aws-user-personalization-v2
- Created:** September 12, 2026, 05:39 PM (UTC +00:00)
- Solution ARN:** arn:aws:personalize:us-east-2:548722509238:solution/drink-solution-cnielson
- Event type:** -
- Latest solution version ARN:** arn:aws:personalize:us-east-2:548722509238:solution/drink-solution-cnielson/c60ce9ae
- Status:** Active

The 'Training configuration' section shows the 'Solution versions (1)' table, which lists the solution version 'c60ce9ae' as 'Active' with a 'Manual' training type, created on 'September 12, 2026, 13:43 (UTC-4:00)'. Buttons for 'Stop creation' and 'Create solution version' are present.

A blue banner at the bottom of the console introduces 'user deletion' capabilities, stating: 'You can now delete all references to specific users from an Amazon Personalize dataset group. This makes it easier to maintain GDPR compliance and manage your data.' A 'Create data deletion job' button is also visible.

Below the console, a 'CloudShell' terminal window is open, showing the command to describe the solution version and its output:

```
$ aws personalize describe-solution-version --solution-version-arn "arn:aws:personalize:us-east-2:548722509238:solution/drink-solution-cnielson/c60ce9ae"
```

```
{
  "solutionVersion": {
    "name": "drink-solution-cnielson/c60ce9ae",
    "solutionVersionArn": "arn:aws:personalize:us-east-2:548722509238:solution/drink-solution-cnielson/c60ce9ae",
    "solutionArn": "arn:aws:personalize:us-east-2:548722509238:solution/drink-solution-cnielson",
    "performMLO": false,
    "recipeArn": "arn:aws:personalize::recipe:aws-user-personalization-v2",
    "datasetGroupArn": "arn:aws:personalize:us-east-2:548722509238:dataset-group/dtsc-671-drink-recs-cnielson",
    "solutionConfig": {
      "autoTrainingConfig": {
        "schedulingExpression": "rate(7 days)"
      }
    },
    "trainingHours": 0.541,
    "trainingMode": "FULL",
    "status": "ACTIVE",
    "creationDateTime": "2026-09-12T17:43:06.822000+00:00",
    "lastUpdatedDateTime": "2026-09-12T18:08:57.959000+00:00",
    "trainingType": "MANUAL"
  }
}
```

3) [50 points] In the space below, provide a screenshot of the test results from the campaign (i.e., the list of recommendations) for customer #80 (i.e., CUST_0080).

drink-campaign-cnielson

Campaign inference

To get recommendations for this campaign in your application, use the `getRecommendations` API call. You can learn more about the usage and requirements for this API call in the documentation and the other links listed below.

- [Amazon Personalize GetRecommendations Developer Guide](#)

Campaign ARN [Info](#)
arn:aws:personalize:us-east-2:548722509238:campaign/drink-campaign-cnielson

Test campaign results

Before you use your campaign in your application, you can view the recommendations it generates here.

User ID [Info](#)
This is the user ID of the user you want to see campaign results for. This user ID needs to be obtained from your item-interactions or user dataset.
CUST_0080

Filter name - optional
Choose an existing filter to apply to your user segments or create a new filter.
None

Recommendations (25) [Info](#)
Recommendations include up to 25 items or actions. Any promoted items are distributed randomly. For some custom recipes, the reason column lists if the item is included through exploration or as a popular item placeholder.

Recommendation ID
RID-08-4a24-8714-39e54b3e4a65-CID-26c7f3

Item ID	Score	Recommendation reason
Pink Drink	0.0734805	-
Caffe Americano	0.0646630	-
Iced Brown Sugar Oatmilk Shaken Espresso	0.0633124	-
Vanilla Bean Frappuccino	0.0576198	-
Caffe Mocha	0.0513455	-
Flat White	0.0472927	-
Cold Brew	0.0454811	-
Iced Pumpkin Spice Latte	0.0418329	-
Matcha Green Tea Latte	0.0402704	-
Espresso	0.0389545	-

```
~ $ aws personalize-runtime get-recommendations --campaign-arn "arn:aws:personalize:us-east-2:548722509238:campaign/drink-campaign-cnielson" --user-id "CUST_0080"
```

```
{
  "itemList": [
    {
      "itemId": "Pink Drink",
      "score": 0.0734805
    },
    {
      "itemId": "Caffe Americano",
      "score": 0.064663
    },
    {
      "itemId": "Iced Brown Sugar Oatmilk Shaken Espresso",
      "score": 0.0633124
    },
    {
      "itemId": "Vanilla Bean Frappuccino",
      "score": 0.0576198
    },
    {
      "itemId": "Caffe Mocha",
      "score": 0.0513455
    },
    {
      "itemId": "Flat White",
      "score": 0.0472927
    },
    {
      "itemId": "Cold Brew",
      "score": 0.0454811
    },
    {
      "itemId": "Iced Pumpkin Spice Latte",
      "score": 0.0418329
    },
    {
      "itemId": "Matcha Green Tea Latte",
      "score": 0.0402704
    },
    {
      "itemId": "Espresso",
      "score": 0.0389545
    }
  ]
}
```